

## Quiz 1: Solutions

Show your reasoning and give exact answers. Do not use electronic tools.

1. Let  $f(x) = \sqrt{x}$  and  $g(x) = 1/(x - 2)$ , each on its largest real domain. Find  $f \circ g$  and  $g \circ f$ , including their domains. Are these the same function?

**Solution.**

The two compositions are

$$(f \circ g)(x) = \sqrt{\frac{1}{x-2}}, \quad (g \circ f)(x) = \frac{1}{\sqrt{x}-2}.$$

For  $f \circ g$ , we need  $x \neq 2$  and  $1/(x-2) \geq 0$ . The reciprocal is positive exactly when  $x > 2$ , so

$$D_{f \circ g} = (2, \infty).$$

For  $g \circ f$ , we need  $x \geq 0$  and  $\sqrt{x} \neq 2$ , so

$$D_{g \circ f} = [0, 4) \cup (4, \infty).$$

The domains differ, so these are not the same function.

2. Solve each equation (don't forget to check the domain of the original expression):

(a)  $\ln(x+2) + \ln(x-1) = \ln 4$ .

(b)  $2 \ln x = \ln(3x+4)$ .

**Solution.**

(a) The original equation requires  $x+2 > 0$  and  $x-1 > 0$ , hence  $x > 1$ . On this domain, the logarithm laws and one-to-one property give

$$\ln((x+2)(x-1)) = \ln 4 \iff (x+2)(x-1) = 4.$$

Thus

$$x^2 + x - 6 = (x+3)(x-2) = 0.$$

Of the candidates  $-3$  and  $2$ , only  $2$  is in the domain. It satisfies  $\ln 4 + \ln 1 = \ln 4$ . The solution is  $\boxed{x=2}$ .

(b) The original equation requires  $x > 0$  and  $3x+4 > 0$ , hence  $x > 0$ . On this domain,

$$\ln(x^2) = \ln(3x+4) \iff x^2 = 3x+4.$$

Factoring gives  $(x-4)(x+1) = 0$ . Only  $4$  is in the domain, and  $2 \ln 4 = \ln 16 = \ln(3 \cdot 4 + 4)$ . The solution is  $\boxed{x=4}$ .